Unified Military Analytics & Comparison Dashboard

Global Military Analysis System • 2025

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Project Overview

Comprehensive Coverage

Interactive dashboard analyzing military capabilities across 140+ countries with 50+ indicators

Centralized Platform

Unified analytics tool eliminating scattered data sources for defense analysis

Technical Stack

*Utilizes **Python (requests, BeautifulSoup, pandas)** for data scraping and cleaning, and **Power BI** for building interactive dashboards*

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Problem Statement

Current Challenges

- *Military data scattered across multiple sources*
- *Difficult to compare countries systematically*
- *No unified platform for comprehensive analysis*
- *Time-consuming manual data collection*

Need for centralized dashboard

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Project Objectives

01

Data Scraping

Automated web scraping from GlobalFirepower database

02

Data Engineering

Clean, structure, and validate dataset quality

03

KPI Generation

Create standardized Key Performance Indicators

04

Dashboard Development

Build interactive visualization interface

05

Testing & Validation

Ensure accuracy and usability

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Data Collection & Technical Stack

Data Sources

*Primary source: **GlobalFirepower** database*

URLs programmatically generated from dataset file

Core Technologies

- ***Python 3.x** - Main programming language*
- ***BeautifulSoup** - Web scraping library*
- ***Pandas** - Data manipulation & analysis*
- ***Matplotlib/Seaborn** - Visualizations*

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Key Metrics Used

Engineered metrics enabling standardized military power comparison

1

Total military personnel

Total active and reserve forces available for defense

2

Aircraft

Total number of fighter jets, transport, and support aircraft

3

Defense budget

Total military spending indicating financial strength

4

Economic indicators

Includes GDP, reserves, and resources impacting military capability

5

Naval Power

Number of ships, submarines, and overall naval fleet strength

6

Land Power (Tanks & Vehicles)

Strength of ground forces including tanks and armored vehicles

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KPI Engineering

1

Power Index Rank Gap

Difference in ranking between countries to measure relative strength

2

Assets per Capita

Military assets available per population

3

Budget-to-GDP Ratio

Percentage of GDP spent on defense

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System Architecture & Workflow

Web Scraping

BeautifulSoup data extraction

Data Cleaning

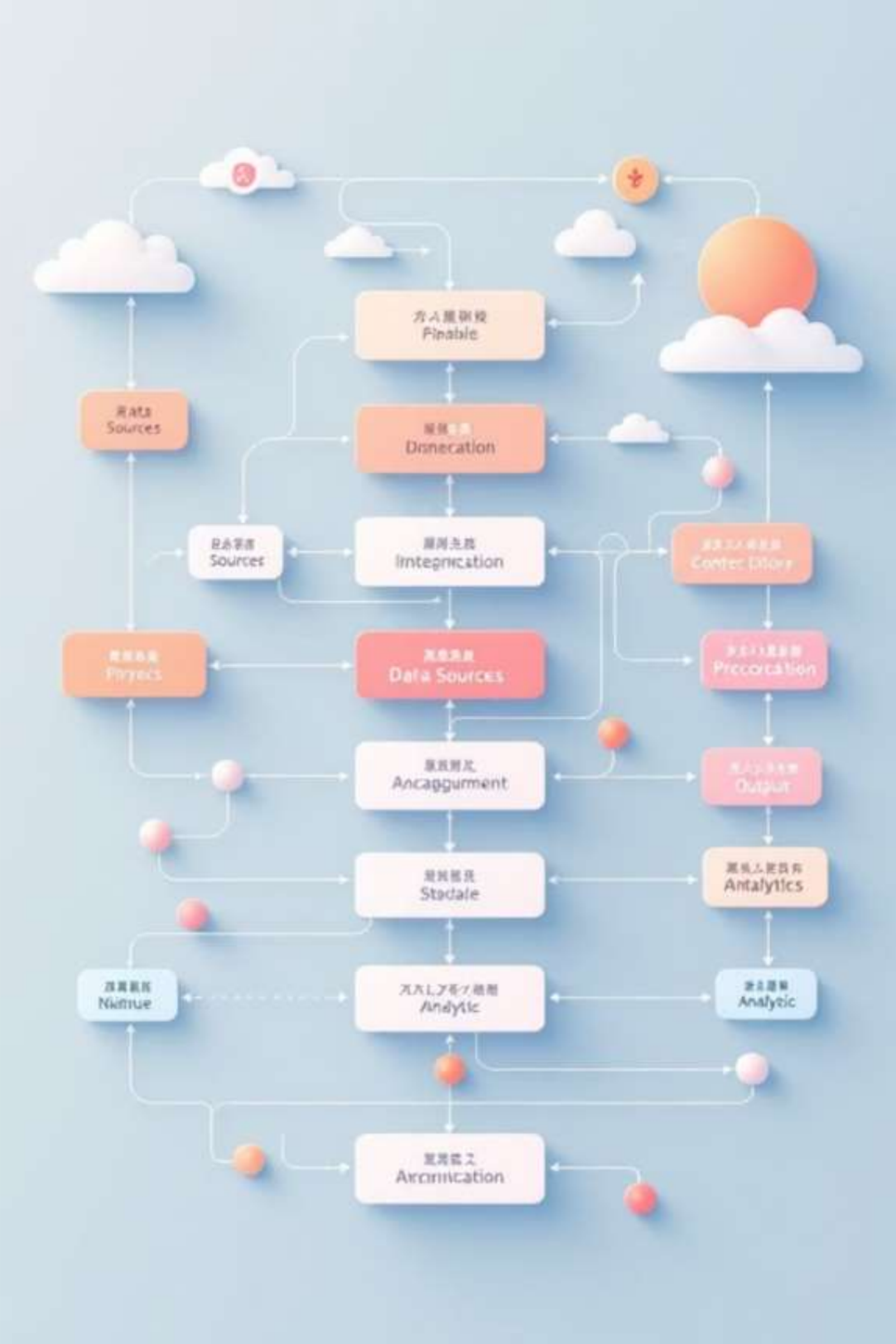
Pandas transformation

KPI Engineering

Metric calculation

Dashboard

Interactive visualization



Dashboard Module Architecture

Quick Stats

High-level military indicators at a glance

Nation Overview

Comprehensive country-specific military profiles

Compare Powers

Side-by-side analysis of multiple nations

Coalition Builder

Strategic alliance capability assessment

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Dashboard Functionality

- *Analyze global military data in one place*
- *Compare countries and rankings*
- *Visualize key metrics using Power BI*
- *Evaluate alliances and trends*

Dashboard Interface & Visualizations

Key Features

- *Asset distribution charts*
- *Budget trend visualizations*
- *Interactive world map*
- *Global ranking display*

Interactive Elements

- *Region filters*
- *Country selectors*
- *Time period controls*
- *Metric comparisons*

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Conclusion & Impact

Unified Platform

Centralized military analytics eliminating data fragmentation

Enhanced Insights

Standardized comparison enabling strategic analysis

Scalable Design

Architecture supports future expansion and additional metrics

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Thank You